

Machine Learning for Natural Language Understanding

Prof. Achim Rettinger – Master NLP

Sources: Lecture and Exercise

Key Content

- Llama 3 (Section 4.1 and 4.2)
 - <https://ai.meta.com/research/publications/the-llama-3-herd-of-models/>
- DeepSeek-V3 (Section 5)
 - <https://arxiv.org/pdf/2412.19437>
- Instruct-GPT
 - <https://arxiv.org/pdf/2203.02155.pdf>

Further Reading

- NLU
 - https://youtu.be/tZ_Jrc_nRJY
(up to minute 51)

Traditional Tasks of NLU

- Relation Extraction
 - https://youtu.be/pO3Jsr31s_Q?t=445
- Grounding
 - https://youtu.be/7b2_3dDTKMc?t=670
- Semantic Parsing
 - <https://youtu.be/C5bdf1sg7rs>
- Question Answering
 - https://youtu.be/NcqfHa0_YmU
- Sentiment Analysis
 - <https://youtu.be/O1Xh3H1uEYY?t=241>
(up to Minute 41)

Flipped Classroom Questions

- How does Post Training differ from Unsupervised Pretraining?
- Name examples for typical NLU benchmarking tasks.
- What is the difference between Supervised Finetuning Data and Reinforcement Learning?
- Name different ways to obtain Supervised Finetuning data.



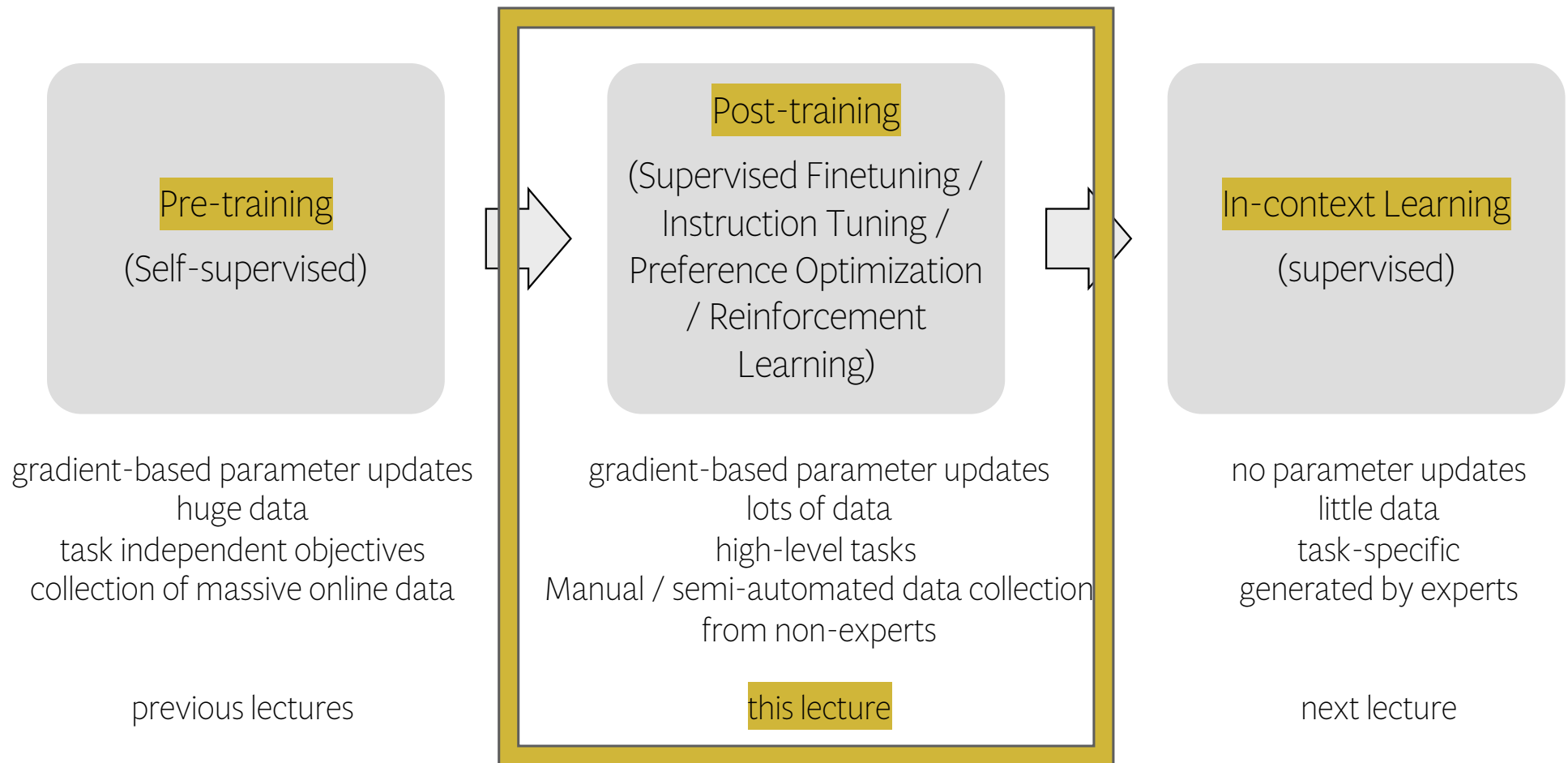
Post Training

Lecture 9

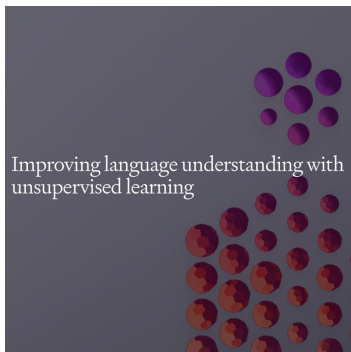
Recap: Training Steps of LLMs

Chapter 8.1

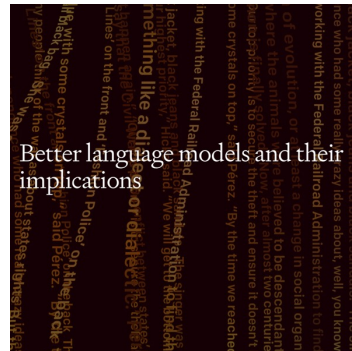
Recap: Training Steps of Instruction-tuned LLMs



Recap: Evolution of GPT



- 2018
- 12 layers x 12 heads
- Input context: 512
- Zero-shot performance



- 2019
- 48 layers x 48 heads
- Input context : 1024
- Task conditioning

Language models are few-shot learners

[Read paper](#)

[Language GPT-3 Publication Mismatch](#)

Abstract

Recent work has demonstrated substantial gains on many NLP tasks and benchmarks by pre-training on a large corpus of text followed by fine-tuning on a specific task. While typically task-specific in architecture, this method still requires task-specific fine-tuning datasets of thousands or tens of thousands of examples. By contrast, humans can generally perform a new language task from only a few examples or from simple instructions – something which current NLP systems still largely struggle to do. Here we show that scaling up language models greatly improves task-specific, few-shot performance, sometimes even reaching competitiveness with prior state-of-the-art fine-tuning approaches. Specifically, we train GPT-3, an autoregressive language model with 175 billion parameters, 10x more than any previous non-sparse language model, and test its performance in the few-shot setting. For all tasks, GPT-3 is applied without any gradient updates or fine-tuning, with tasks and few-shot demonstrations specified purely via text interaction with the model. GPT-3 achieves strong performance on many NLP datasets, including translation, question answering, and short tasks, as well as several tasks that require on-the-fly reasoning or domain adaptation, such as unscrambling words, using a novel word in a sentence, or performing 3-digit arithmetic. At the same time, we also identify some datasets where GPT-3's few-shot learning still struggles, as well as some datasets where GPT-3 faces methodological issues related to training on large web corpora. Finally, we find that GPT-3 can generate samples of news articles which human evaluators have difficulty distinguishing from articles written by humans. We discuss



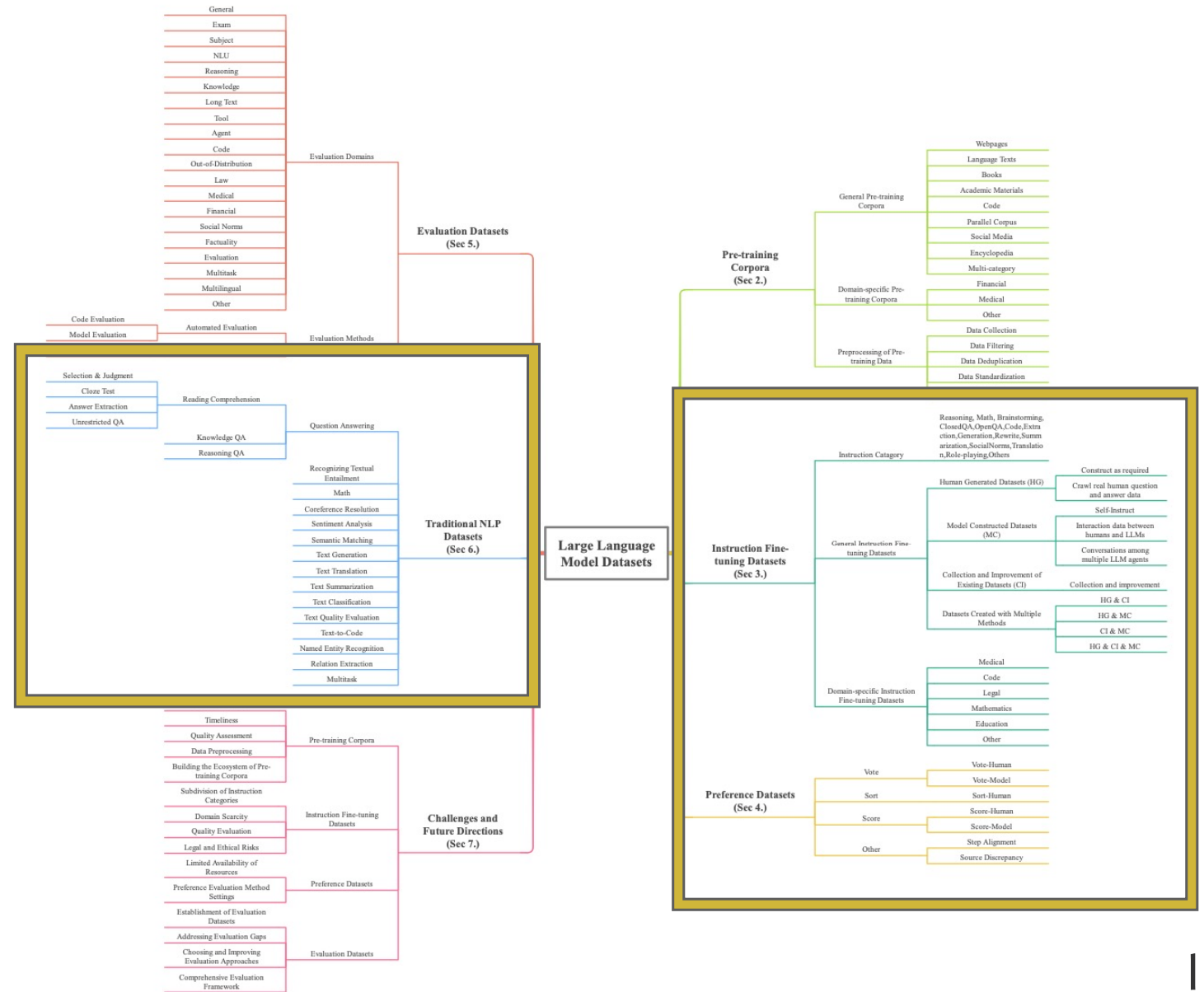
- 2020
- 96 layers x 96 attention
- Input context: 2048
- In-context learning

- 2023
- ???
- Input context : 4096
- ???

- 2022
- Instruct GPT / ChatGPT
- Instruction-following LLM

Recap: Data Sets Overview

(<https://arxiv.org/pdf/2402.18041>)



Two different types of post-training data

- **Supervised Finetuning (SFT) Data**: Input: Instruction, Output: Correct Answer (Section 9.1)
- **Preference Data (PD)**: Input Instruction, Output: Set of potential answers, manual scoring/ranking of answers (Section 9.2)

Different ways of obtaining post-training data

- Take existing data bases or NLP benchmarks and semi-automatically generate SFT data from that (aka “natural instructions”)
- Take a deductive reasoning (rule-based) system (e.g. math solver or code compiler)
- Take an existing Foundation Models and obtain SFT data from it (aka “unnatural instructions”)
- Microwork / crowdsourcing to obtain Preference Data.

Supervised Finetuning (SFT)

Chapter 9.1

Two Training Examples for SFT

```
{  
  "instruction": "Write a limerick about a  
                  pelican.",  
  "input": "",  
  "output": "There once was a pelican so fine,  
             \nHis beak was as colorful as  
sunshine,\nHe would fish all day,\nIn  
a very unique way,\nThis pelican was  
truly divine!\n\n\n",  
},
```

```
{  
  "instruction": "Identify the odd one out from  
                  the group.",  
  "input": "Carrot, Apple, Banana, Grape",  
  "output": "Carrot\n\n",  
},
```




Traditional NLU Benchmark Data

Section 9.1.1

Tasks requiring NLU

Content Analysis

Text Generation

- Document Classification
 - Sentiment Analysis
 - Fake News Detection (determine the truth of statements)
 - Natural Language Inference (judge its entailments)
- Question Answering
 - Conversational Systems
 - Machine Translation
 - Text Summarization



What are examples of common NLU tasks about?

- Relation Extraction

The task of relation extraction

Relation extraction is the task of extracting from natural language text relational triples such as:

```
(founders, SpaceX, Elon_Musk)
(has_spouse, Elon_Musk, Talulah_Riley)
(worked_at, Elon_Musk, Tesla_Motors)
```

- Grounding

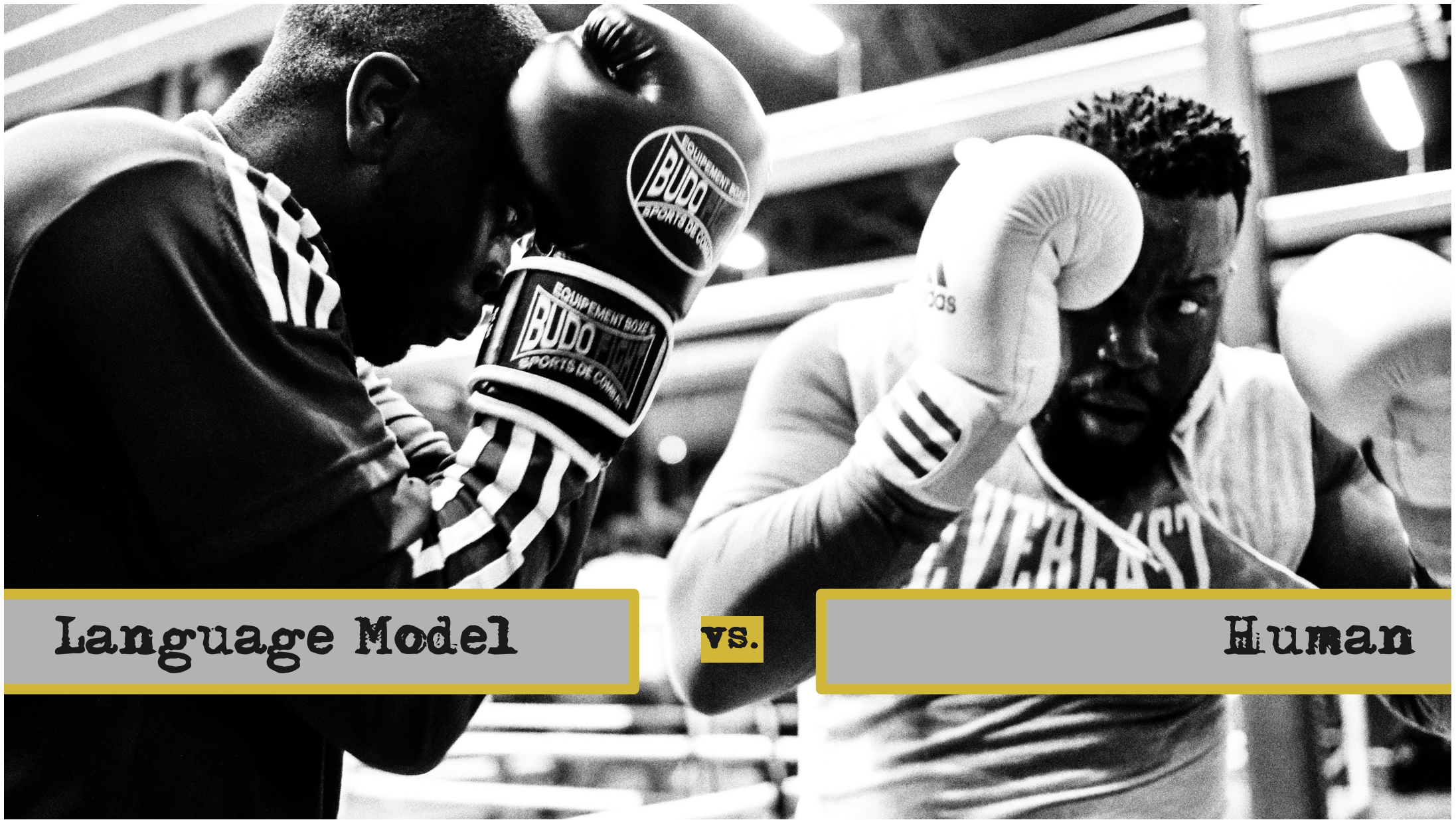
- Language is grounded in a social and physical world

- Semantic Parsing

- E.g., semantic dependency parsing:
Identify semantic relationships between words in a text

- Coreference Resolution

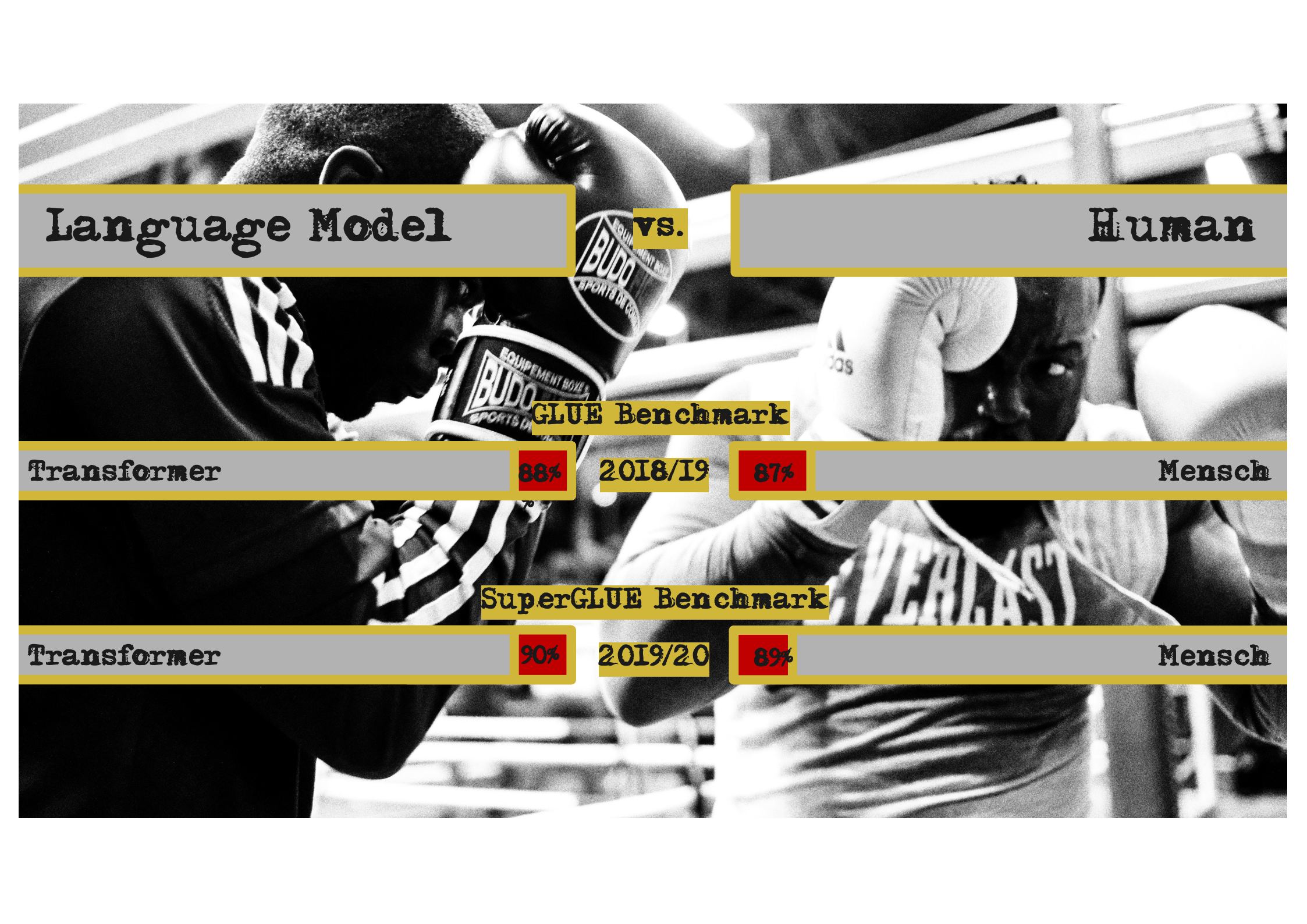
- Identify all mentions in text that refer to the same entity in the world



Language Model

vs.

Human



Language Model

vs.

Human

GLUE Benchmark

Transformer

88%

2018/19

87%

Mensch

SuperGLUE Benchmark

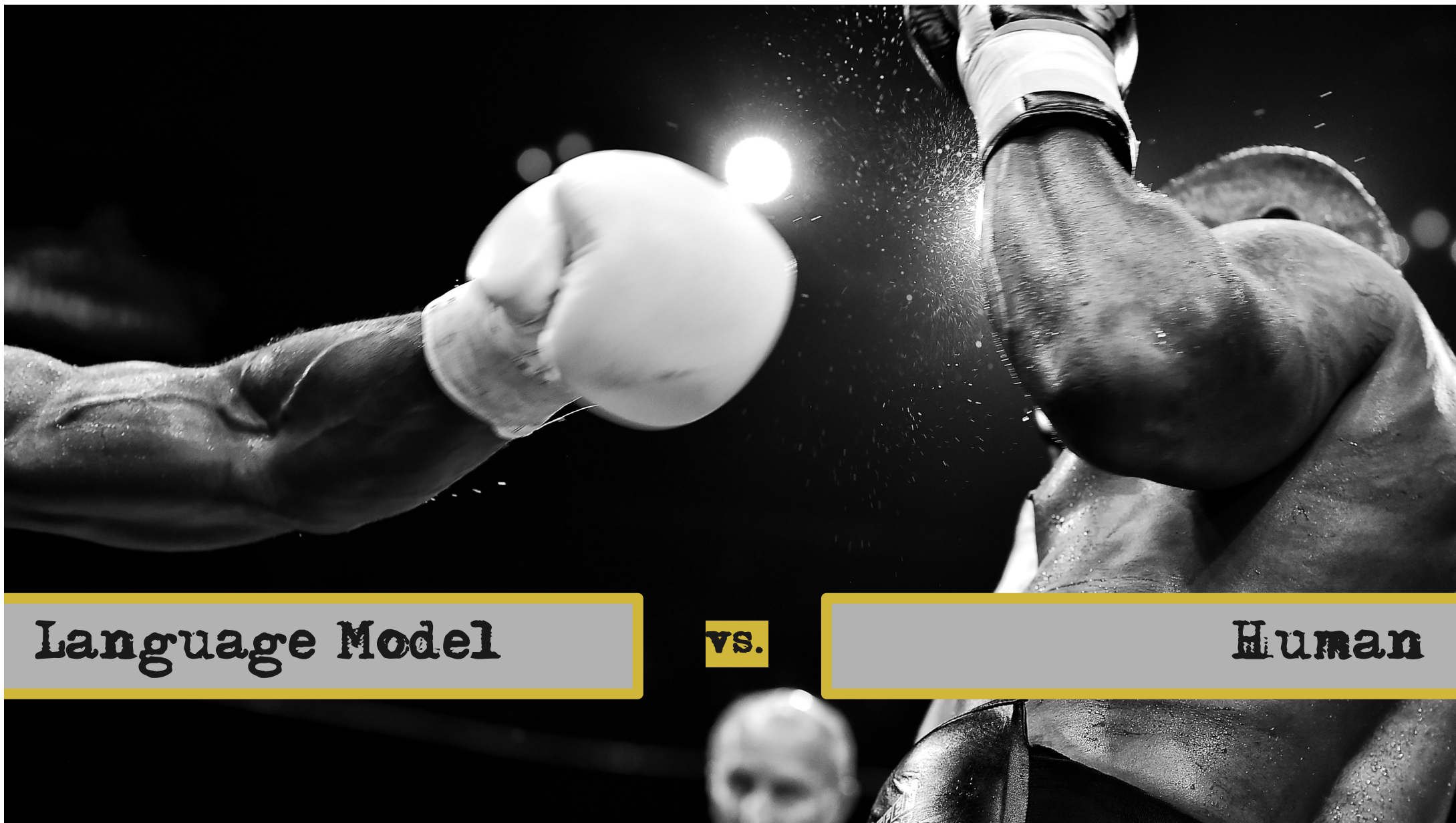
Transformer

90%

2019/20

89%

Mensch



Language Model

vs.

Human

Not updated since 2024

Leaderboard Version: 2.0

Rank	Name	Model	URL	Score	BoolQ	CB	COPA	MultiRC	ReCoRD	RTE	WiC	WSC	AX-b	AX-g
1	JDExplore d-team	Vega v2		91.3	90.5	98.6/99.2	99.4	88.2/62.4	94.4/93.9	96.0	77.4	98.6	-0.4	100.0/50.0
 2	Liam Fedus	ST-MoE-32B		91.2	92.4	96.9/98.0	99.2	89.6/65.8	95.1/94.4	93.5	77.7	96.6	72.3	96.1/94.1
3	Microsoft Alexander v-team	Turing NLR v5		90.9	92.0	95.9/97.6	98.2	88.4/63.0	96.4/95.9	94.1	77.1	97.3	67.8	93.3/95.5
4	ERNIE Team - Baidu	ERNIE 3.0		90.6	91.0	98.6/99.2	97.4	88.6/63.2	94.7/94.2	92.6	77.4	97.3	68.6	92.7/94.7
5	Yi Tay	PaLM 540B		90.4	91.9	94.4/96.0	99.0	88.7/63.6	94.2/93.3	94.1	77.4	95.9	72.9	95.5/90.4
 6	Zirui Wang	T5 + UDG, Single Model (Google Brain)		90.4	91.4	95.8/97.6	98.0	88.3/63.0	94.2/93.5	93.0	77.9	96.6	69.1	92.7/91.9
 7	DeBERTa Team - Microsoft	DeBERTa / TuringNLRv4		90.3	90.4	95.7/97.6	98.4	88.2/63.7	94.5/94.1	93.2	77.5	95.9	66.7	93.3/93.8
8	SuperGLUE Human Baselines	SuperGLUE Human Baselines		89.8	89.0	95.8/98.9	100.0	81.8/51.9	91.7/91.3	93.6	80.0	100.0	76.6	99.3/99.7

A close-up photograph of a metallic surface, possibly a turbine or engine component, featuring a circular opening and a vertical rod. The image is overlaid with a semi-transparent white band containing text.

Scaling SFT Data

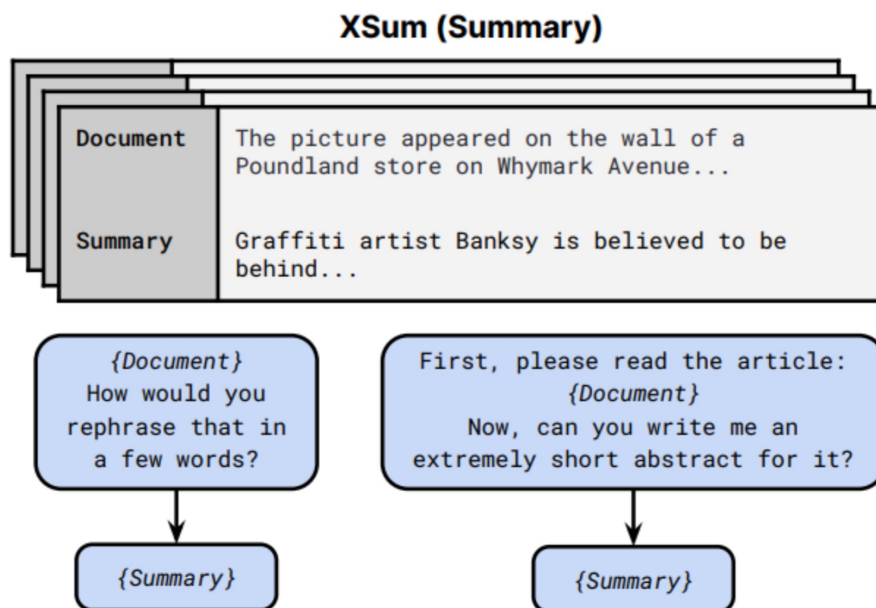
Section 9.1.2

Characteristics of instruction-tuning data

- **Data source:** How was the data obtained?
- **Data quality:** Was any filtering done to improve the quality of the generated data?
- **Domain and language coverage:** Most datasets cover general QA-style use cases and are in English.
- **Number of dialog turns:** A dialog turn is an utterance by one speaker. Most datasets are single-turn.
- **License terms:** Data generated using OpenAI models is subject to the OpenAI terms of use, which prohibit using the data to develop competing models.

Take existing NLU benchmark data and...

- ... add instructions semi-manual
- ... use LLM to generate instructions



question generation (from MC-TACO)

- **Title:** Writing questions that involve commonsense understanding of "event duration".
- **Definition:** In this task, we ask you to write a question that involves "event duration", based on a given sentence. Here, event duration is defined as the understanding of how long events typically last. For example, "brushing teeth", usually takes few minutes.
- **Emphasis & Caution:** The written questions are not required to have a single correct answer.
- **Things to avoid:** Don't create questions which have explicit mentions of answers in text. Instead, it has to be implied from what is given. In other words, we want you to use "instinct" or "common sense".

Positive Example

- **Input:** Sentence: Jack played basketball after school, after which he was very tired.
- **Output:** How long did Jack play basketball?
- **Reason:** the question asks about the duration of an event; therefore it's a temporal event duration question.

Negative Example

- **Input:** Sentence: He spent two hours on his homework.
- **Output:** How long did he do his homework?
- **Reason:** We DO NOT want this question as the answer is directly mentioned in the text.
- **Suggestion:** -

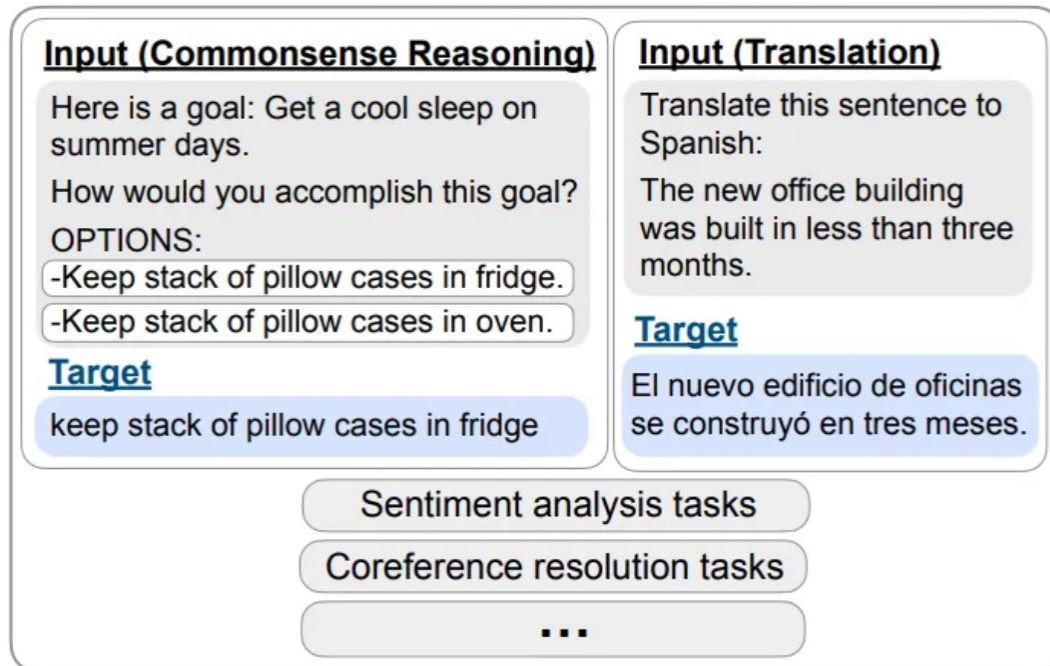
- **Prompt:** Ask a question on "event duration" based on the provided sentence.

Task Instance

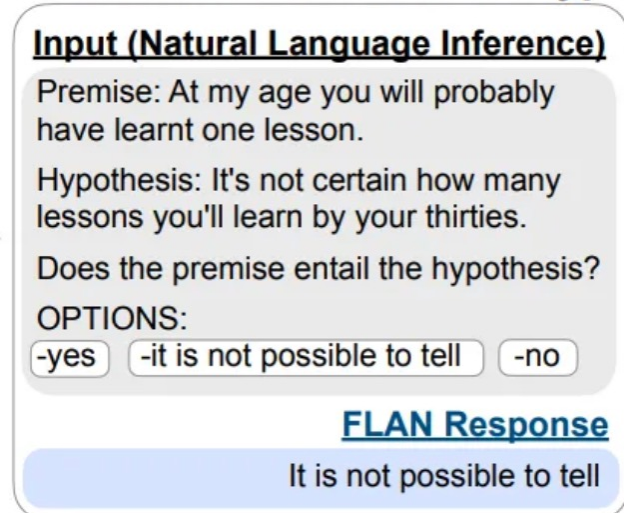
- **Input:** Sentence: Still, Preetam vows to marry Nandini if she meets him again.
- **Expected Output:** How long had they known each other?

Generalizing to unseen tasks (cross-task transfer)

Finetune on many tasks (“instruction-tuning”)



Inference on unseen task type

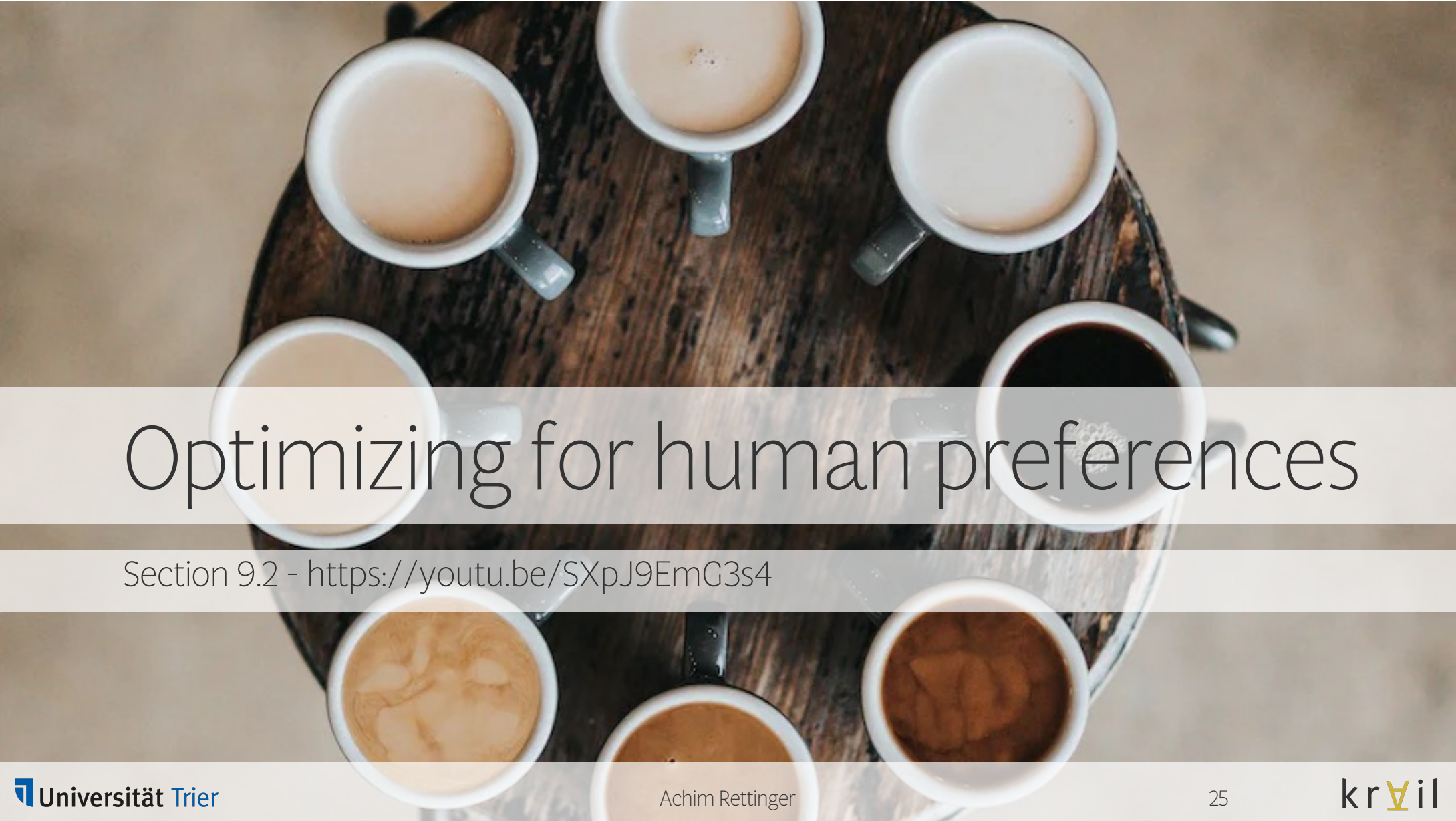


Tasks in the SuperNaturalInstructions dataset



Some typical Lessons Learned

- **Mixing few-shot settings.** Training with mixed zero-shot and few-shot prompts significantly improves performance in both settings.
- **Task diversity.** Large models benefit from continuously increasing the number of tasks.
- **Data augmentation.** Augmenting the data such as by inverting inputs and outputs is beneficial.
- **Quality > quantity.**
- **Imitation != mastery.**
- The **stronger** the base, the better.
- Combination and **diversity wins.**



Optimizing for human preferences

Section 9.2 - <https://youtu.be/SXpJ9EmG3s4>

Key ideas

Measure the quality of a LLM's output by a reward given by a human.

That indicates the human's preference and does not apply to correct vs incorrect answers only.

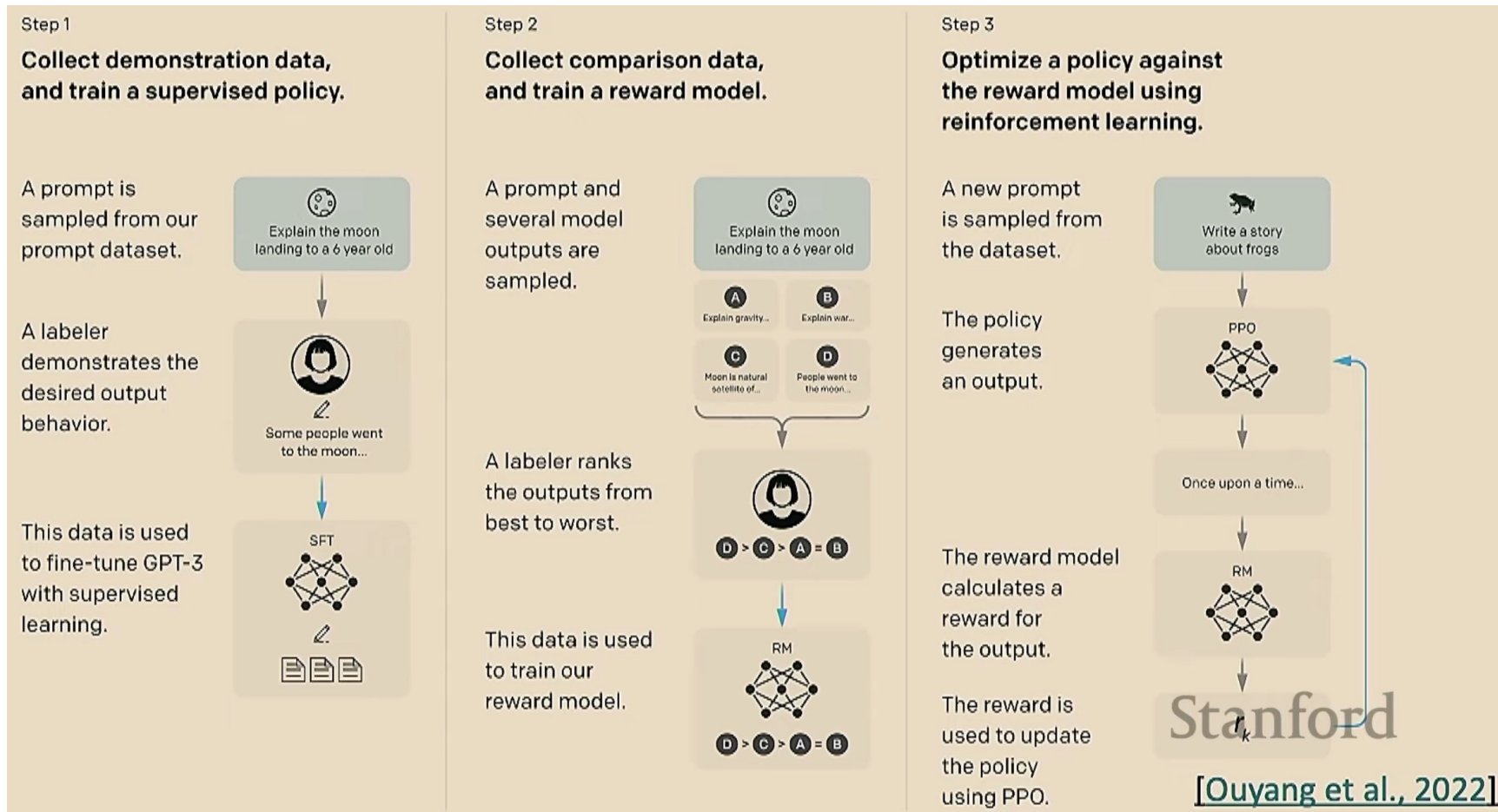
Since humans are not good at giving numerical rewards, ask for pairwise comparisons or ranking of different candidate outputs.

Optimize the model so that it assigns scores that indicate the relative preference of humans between alternative outputs

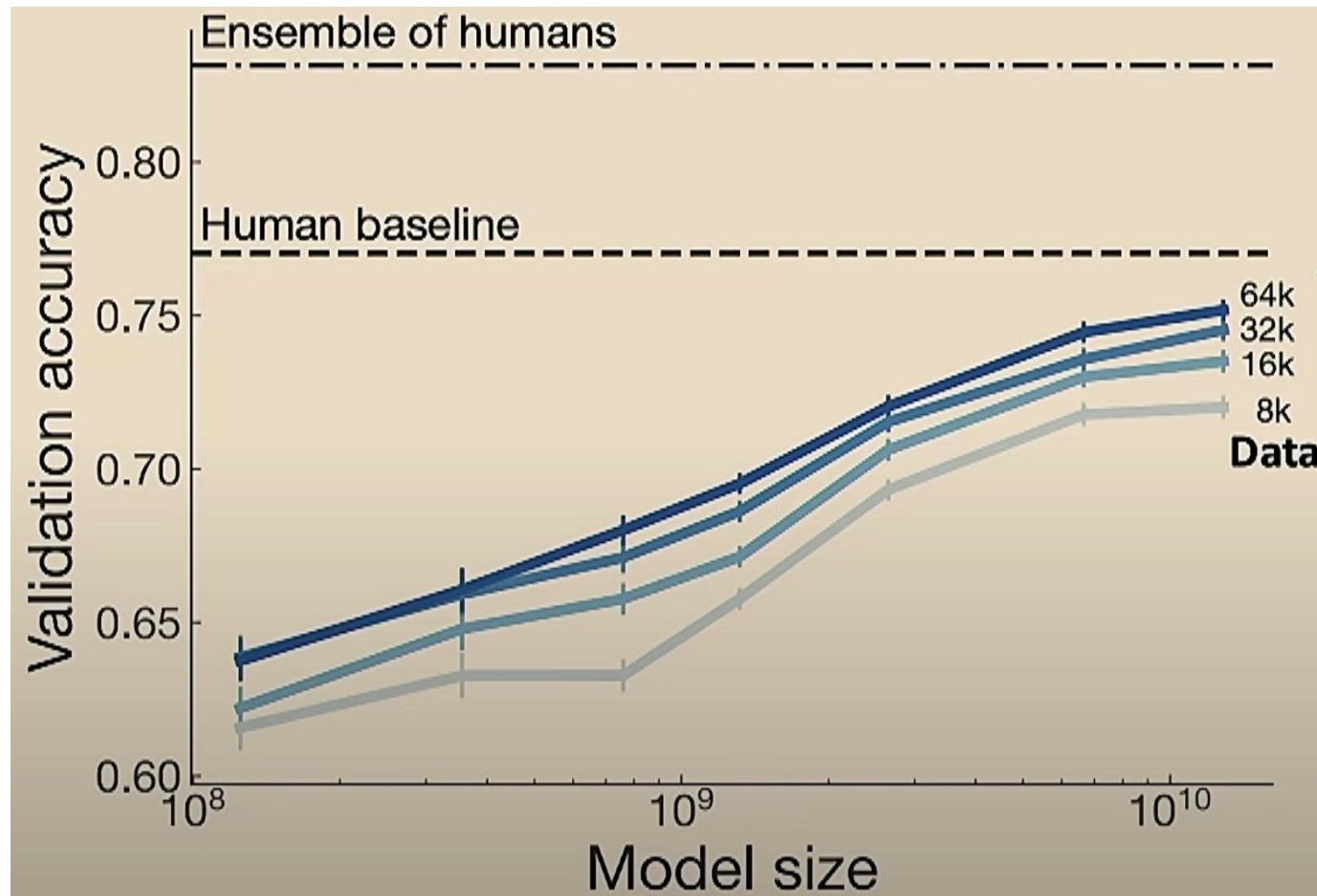
Instruct GPT Post Training (RLHF)

Chapter 9.2.1

RLHF in InstructGPT <https://arxiv.org/pdf/2203.02155.pdf> (Section 1)



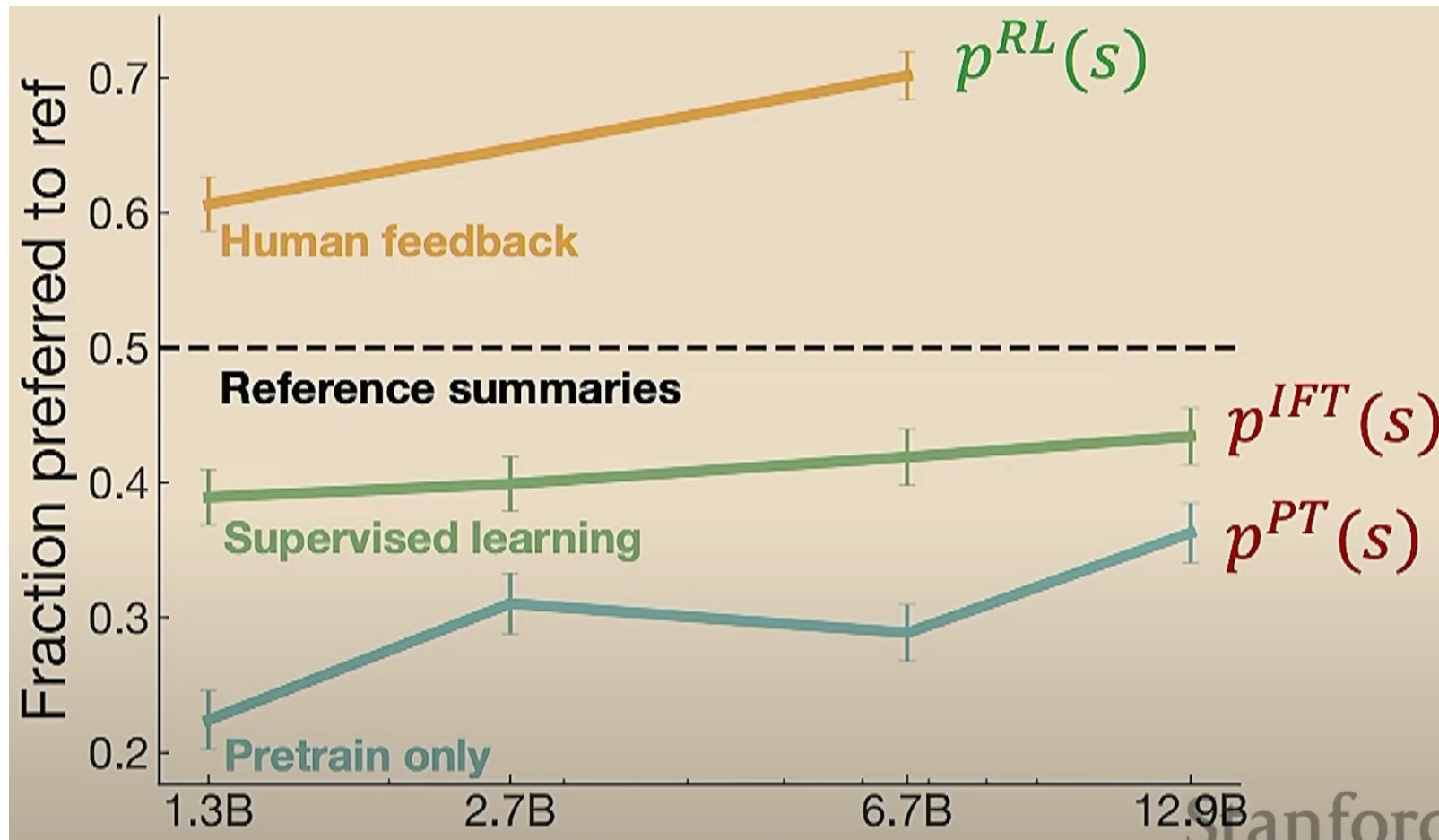
Can a reward model generalize to unseen samples?



Overall steps for Instruction Tuning using RLHF

1. Supervised finetuning: The model is initially trained on a dataset containing instructions and the desired responses. Loss function is based on counting correctness of tokens.
2. Reward modeling: Human raters provide feedback on the model's outputs. This feedback is used to create a reward model, which learns to predict a scalar value of what kinds of outputs are preferred by humans.
3. The RLHF algorithm uses these scores to adjust the LM's parameters to maximize the reward.

Does it work?



Advanced Post-Training

Chapter 9.2.2

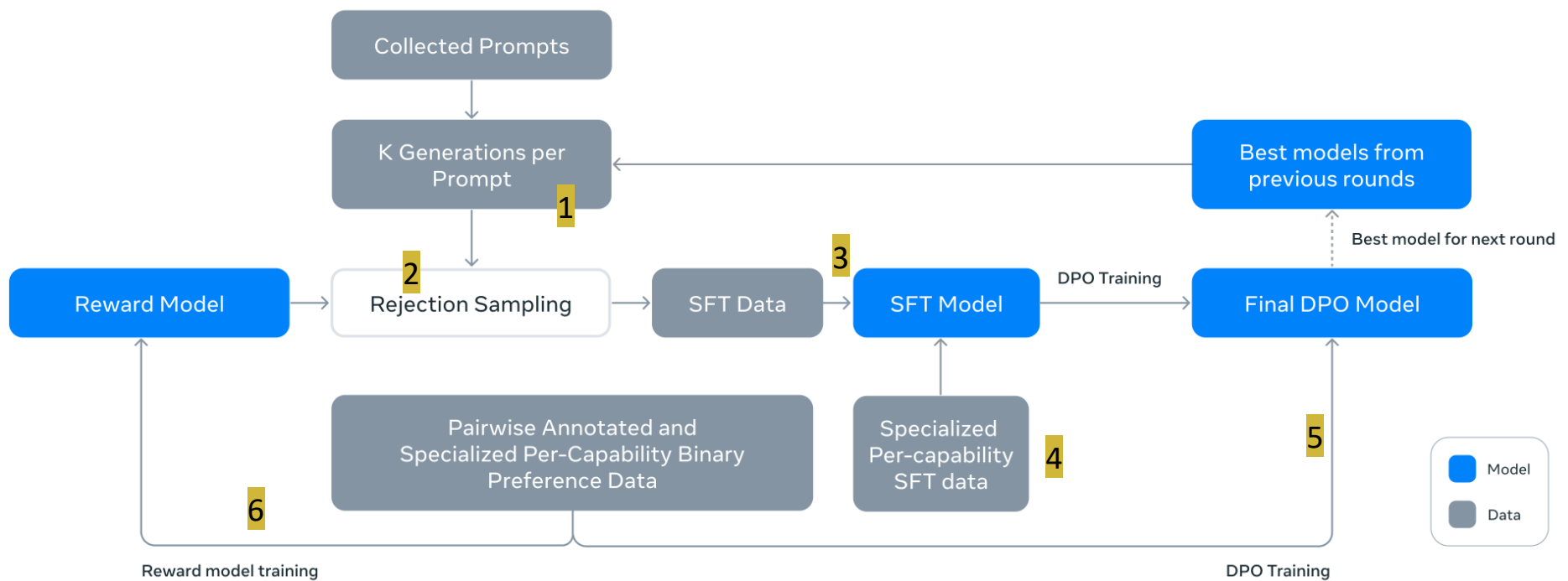
Beyond RLHF

- Process Reward Models (PRMs) T
 - Trained to grade each step of the reasoning, rather than just the final answer. The data format must distinguish between individual steps and their specific correctness labels.
 - This format breaks a single solution into a list of steps, each with a boolean label or score
- Proximal Policy Optimization (PPO).
 - in standard PPO, you need four models (or copies of models) in memory:
 - Policy Model (Actor): The LLM being trained.
 - Reference Model: A frozen copy of the LLM to prevent it from changing too much (KL divergence).
 - Reward Model: Assigns scores (=rewards) to answers.
 - Value Model (Critic): Estimates how much reward the model expects to get from a certain state.
 - In standard PPO, the Advantage (of a specific answer) is calculated as the Reward – Critic's Prediction

Llama 3 Post Training

Chapter 9.2.3

Llama 3 Post Training Approach



Training Process Step-by-Step

1. Use current language models to generate replies given a collection of prompts
2. Use current reward model to select the best candidate
3. Use “Prompt-BestCandidate pairs” for Supervised Finetuning (SFT) of the current language model
4. Use capability-specific data (e.g. from benchmarks) for further SFT of the current language model
5. Use Preference Data for further optimization of the current language model
6. Update Reward Model by asking human annotators to rate the chosen response from step 2.
7. Repeat 6x with updated language/preference model

DeepSeek-V3 Post Training

Chapter 9.2.4

SFT Reasoning Data Generation

- Generated by a DeepSeek-R1 model: The goal is to teach the model how to structure a reasoning trace. The data typically includes a prompt and a response where the reasoning is explicitly written out.
- Key Feature: The `<think>` and `</think>` tags. The model learns to open this tag immediately after the user prompt, generate tokens for "thinking," and close it before giving the final answer
- two distinct types of SFT samples for each instance:
 - the first couples the problem with its original response in the format of `<problem, original response>`
 - the second incorporates a system prompt alongside the problem and the R1 response in the format of `<system prompt, problem, R1 response>`
- fine-tune DeepSeek-V3-Base for two epochs using the SFT dataset

Reinforcement Learning

- Rule-based Reward Model (RM) and a model-based RM
- The data is "unlabelled" in the traditional sense—it consists of prompts and ground-truth answers used to verify the model's self-generated traces.
- Rule-based: The model generates a response (with deductive reasoning). The system executes the code or checks the math.
- Model-based: For questions with free-form ground-truth answers, a reward model is trained from the DeepSeek-V3 SFT checkpoints. Next Group Relative Policy Optimization (GRPO):
 - Group Sampling: For every question (prompt), the model generates a group of distinct outputs (e.g., 64 different responses).
 - Scoring: Each outcome is scored by the RMs
 - Relative Advantage: GRPO calculates the mean and standard deviation of the rewards within that group.
 - If an output scored higher than the group average, encourage this behavior.
 - If it scored lower than average, it has a negative advantage.
 - Optimization: The model is updated to maximize the likelihood of the high-scoring outputs relative to the group.

“The future of language is about the development of language models...”¹



1

Computerlinguistik | Universität Trier | cl.cldh.uni-trier.de